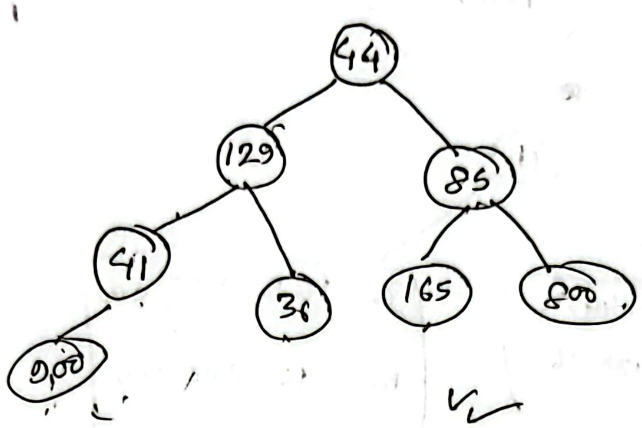
Spring - 22

① x = 41 , y = 44, z = 85, p = 129, r = 36, t = 165

u = 800, v = 900

y p z x r t u v



In → L - Root - right

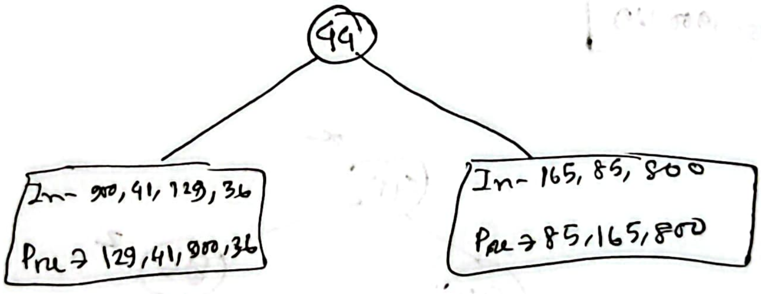
Pre → root - Left → right

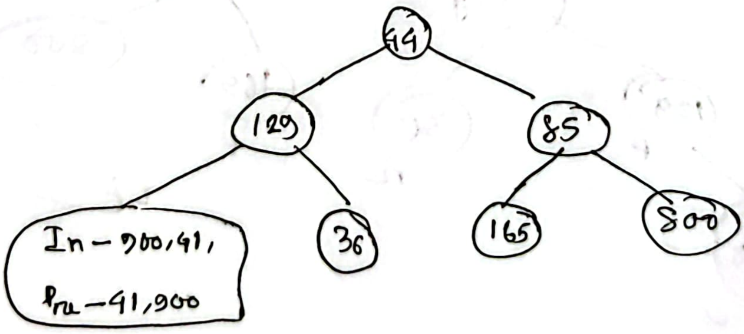
Po → Left → right - root

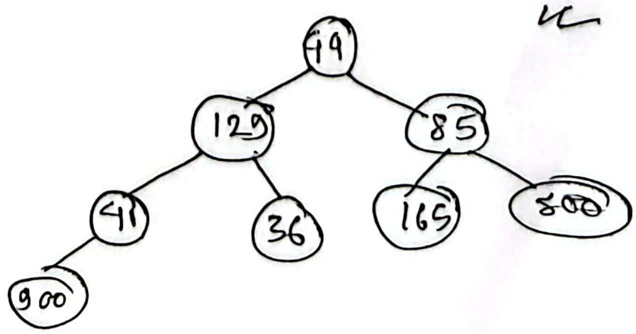
In → 900 → 41 → 129 → 36 → 44 → 165 → 85 → 800

Pre → 44 → 129 → 41 → 900 → 36 → 85 → 165 → 800

level → 44 → 129 → 85 → 41 → 36 → 165 → 800 → 900





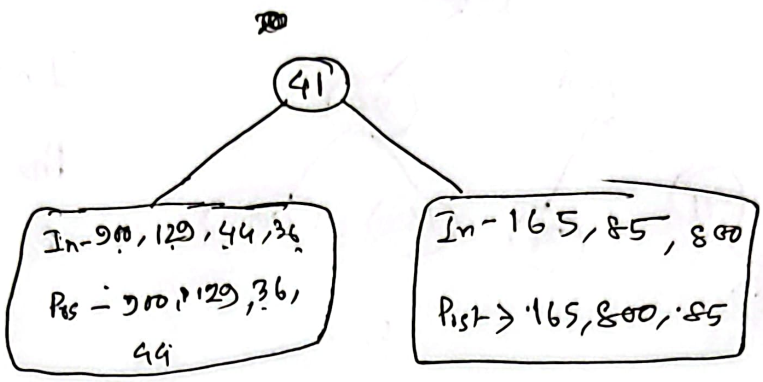


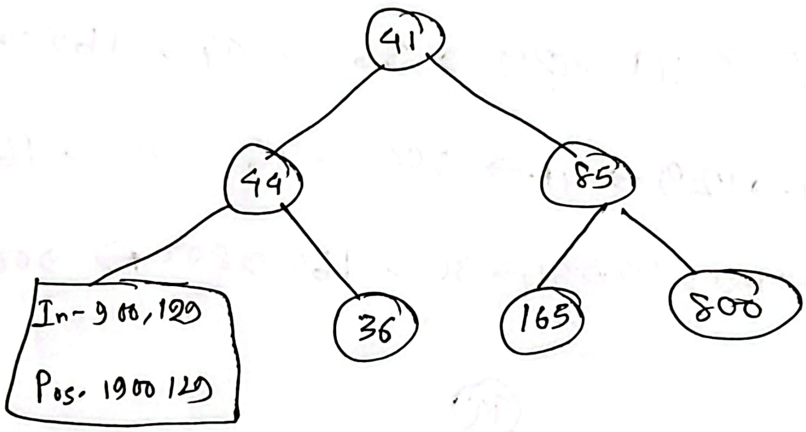
(e)

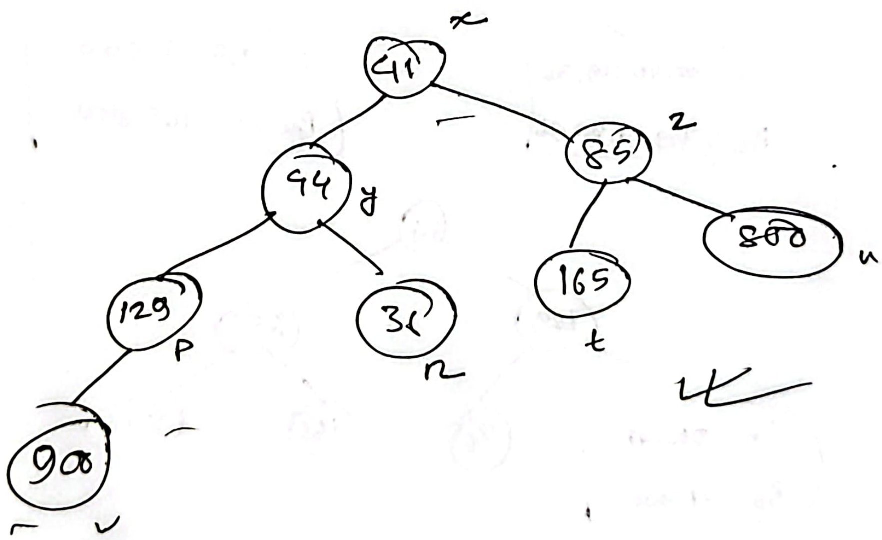
$x = 41, y = 44 \quad z = 85 \cdot P = 129 \quad r = 36 \quad t = 165$ L–Ri–Root

$u = 800, \quad v = 900$

$\boxed{\begin{array}{l} In \to v\ p\ y\ r\ x\ t\ z\ u \\ Post \to v\ p\ r\ y\ t\ u\ z\ x \end{array}}$



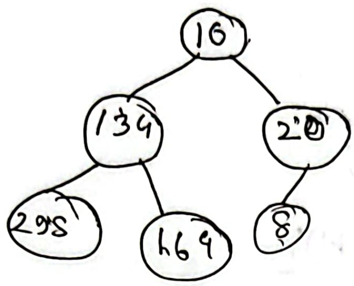




164

(b) 10 x 20 2 y 8

10 134 20 298 164 8



max heap

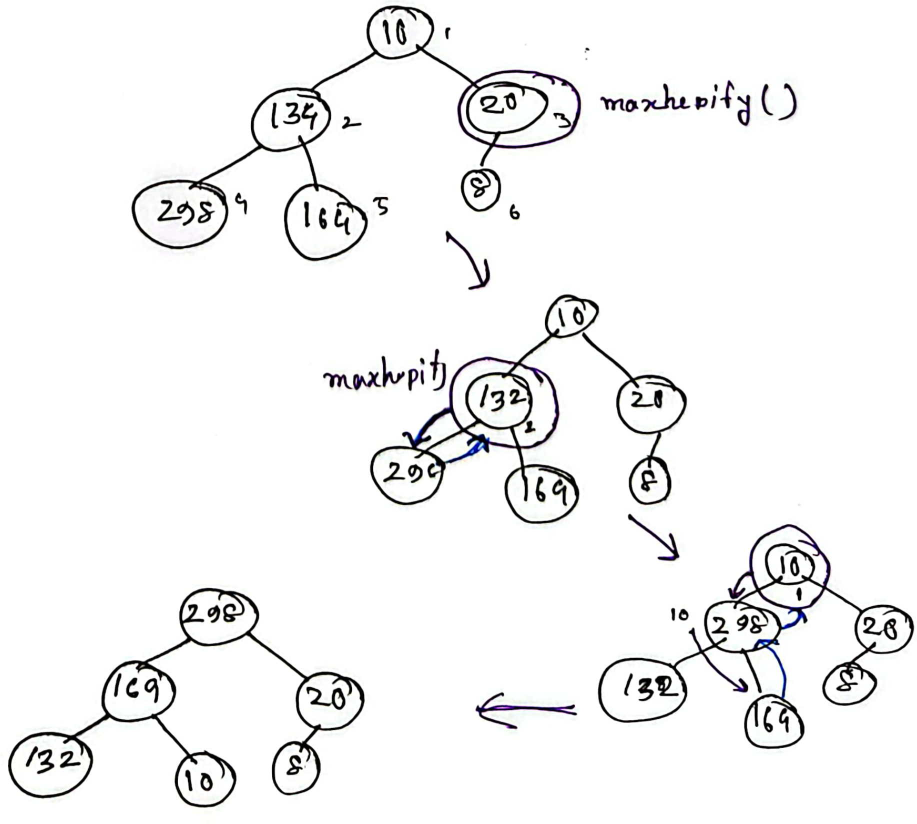
BMH (A)

A. hepSize = A. length

for (i = $\lfloor A.length/2 \rfloor$. downto 1)

$\lfloor \frac{6}{2} \rfloor = 3$

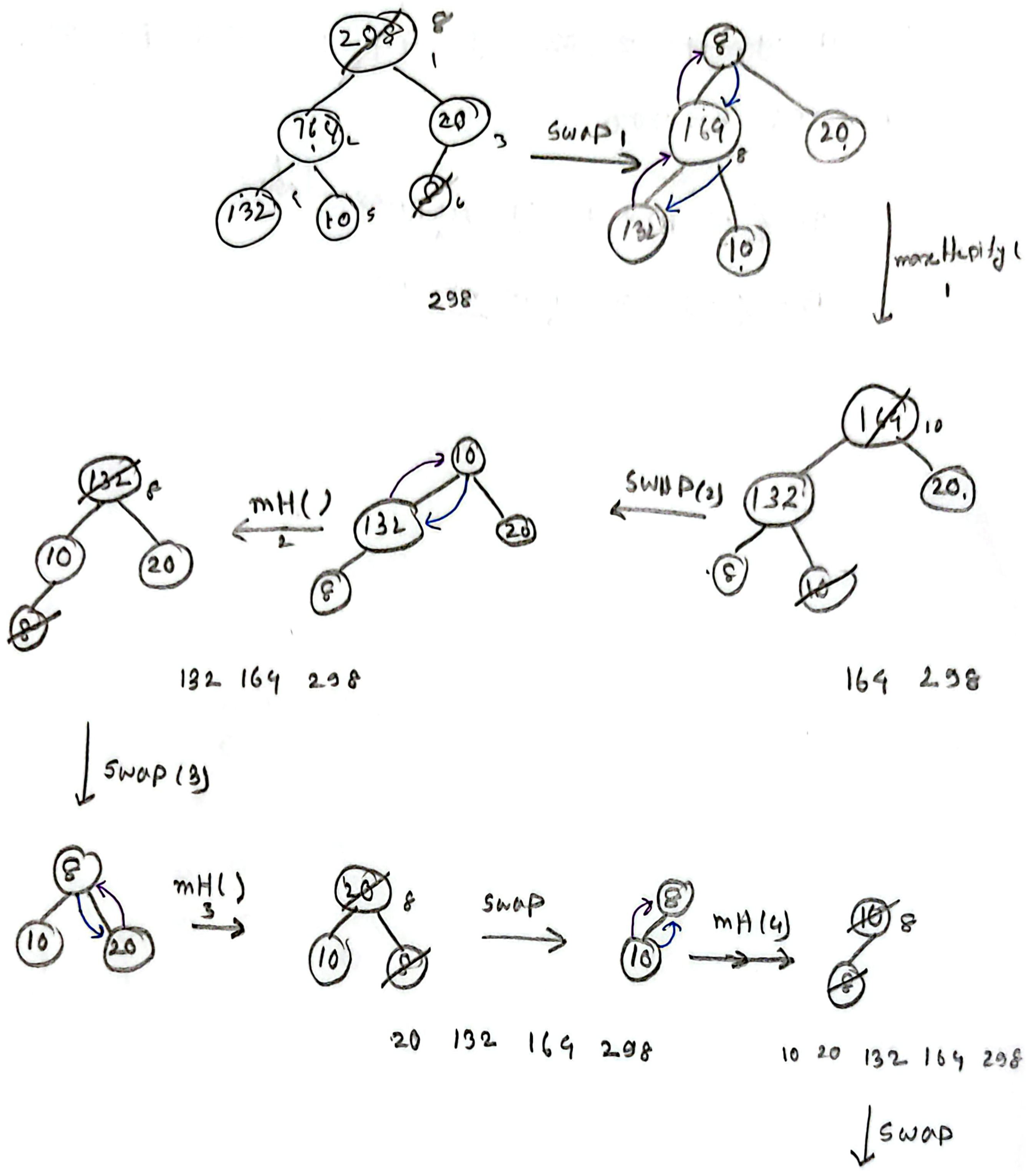
Maxheapify (A, i)



(d)

```
void prieorder(node *root) {

    if (root == NULL)
    {
        return root;
    }

    else
    {
        cout << root -> key << " "

        preOrder(root -> left);

        PreeOrder(root -> right);

    }
}
```

(2)



Finally

8 10 20 132 164 298

(c) $x = 41$, ~~$y = 41$, $z = 42$~~, $y = 44$, $z = 85$, $P = 129$,

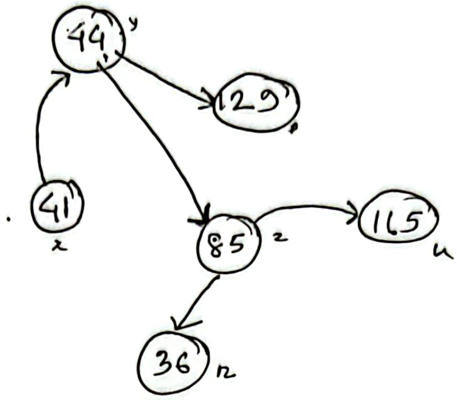
$r = 36$, $t = 900$

$A = \{44, 129, 85, 41\}$ representative

$B = \{36, 900\}$ → repsentative

③ $x = 41, y = 44, z = 85, p = 129, r = 36, u = 165$

$\acute{y}\ \acute{p}\ \acute{z}\ x\ r\ u$



A	y	p	z	x	r	u
y	0	1	1	0	0	0
p	0	0	0	0	0	0
z	0	0	0	0	1	1
x	1	0	0	0	0	0
r	0	0	0	0	0	0
u	0	0	0	0	0	0

44 $\boxed{44} \rightarrow \boxed{29} \rightarrow \boxed{85} \rightarrow$ NULL

29 NULL

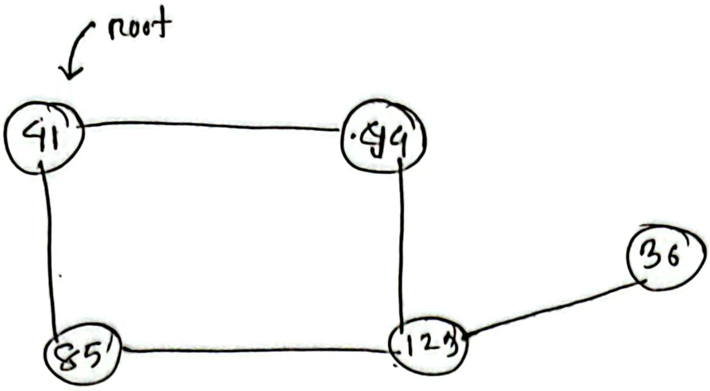
85 $\boxed{85} \rightarrow \boxed{36} \rightarrow \boxed{165} \rightarrow$ NULL

41 $\boxed{41} \rightarrow \boxed{44} \rightarrow$ NULL

36 NULL

165 NULL

(a)



a $\boxed{\cancel{41}, \cancel{44}, \cancel{85}; \cancel{129}, \cancel{36}}$

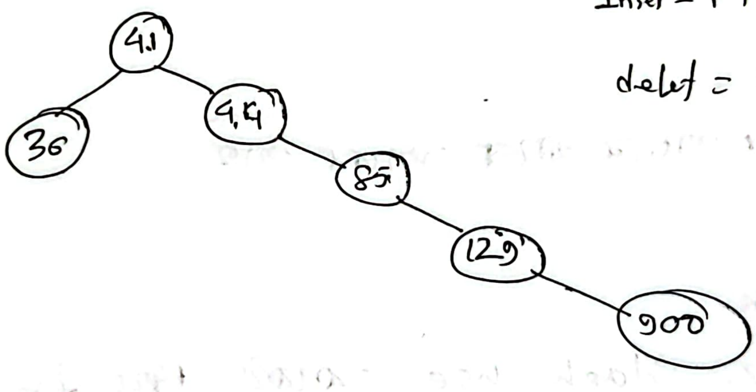
res → 41, 44, 129, 36, 85

(b)

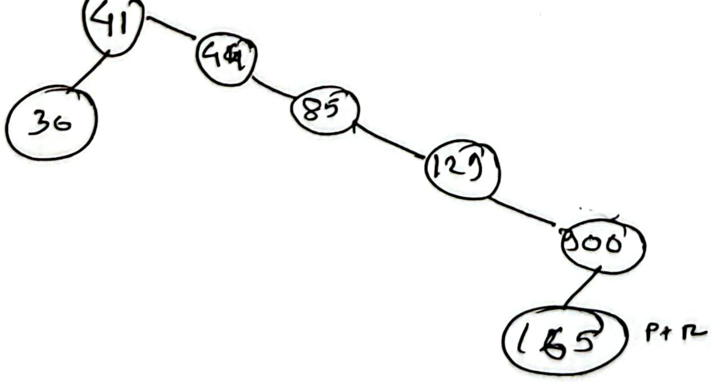
$x = 41$, $y = 44$, $z = 85$, $p = 129$, $r = 36$, $t = 900$

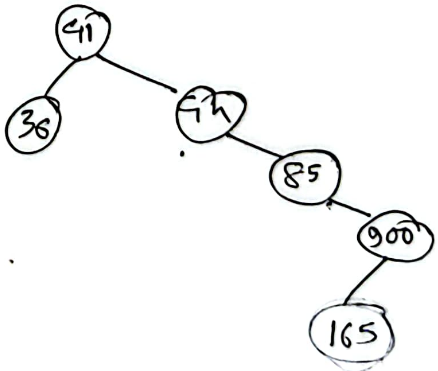
Insert = p + r = 165

delet = 129



Insert





(e)

(d) (i) Graph

nodes (cities), distance (one nodes to another nodes)

(ii) Queue

আগে আসলে আগে পাবেন তাই

(iii) Stack

ez Stack use করে Pre, In Post করি